TECHNICAL IMPLEMENTATION REPORT

Active Directory & Group Policy Lab

Windows Server 2022 | corp.lab | Domain Services | File Services | Workstation Controls



Prepared: 04 November 2026

Lab implementation record with configuration evidence and security review

1. Executive Summary

This project implements a small Windows domain environment using Windows Server 2022, Active Directory Domain Services, DNS, Group Policy, and SMB file sharing. The design separates users by department, assigns department security groups, maps department folders to drive letters, and applies domain account and workstation restrictions.

Project scope Domain: corp.lab | NetBIOS: CORP | Domain controller / file server label: DC01 | Domain client: CLIENT01

Delivered capabilities

- Created the corp.lab forest and verified the domain structure in Active Directory Users and Computers.
- Created Management, IT, HR, Finance, and Workstations OUs with representative users and security groups.
- Created the Corp Lab Data SMB share and four department folders.
- Configured department-specific drive mappings for H:, F:, I:, and M:.
- Configured domain password and account-lockout policies through the Default Domain Policy.
- Configured central workstation restrictions and an HR-specific desktop restriction policy.
- Captured client evidence for the sara.ahmed logon, HR H: drive, Control Panel restriction message, and CD/DVD access denial.

Evidence standard This report distinguishes policy configuration screens from observed client-side test results. A configured policy is not marked as passed unless client behavior is shown in the supplied evidence.

2. Environment and Logical Architecture

Component	Recorded value
Server platform	Windows Server 2022
Domain	corp.lab
NetBIOS	CORP
Domain controller / file server label	DC01
Domain client	CLIENT01
Share local path	C:\SHARED FOLDER\Corp Lab Data
Share UNC path	\\DC01\Corp Lab Data

The lab network was adjusted during troubleshooting. Because the supplied captures contain more than one network state, this report intentionally does not publish a fixed IPv4 address. Confirm the current IP, gateway, DNS server, and VMware network mode in the running VMs before reproducing the lab.

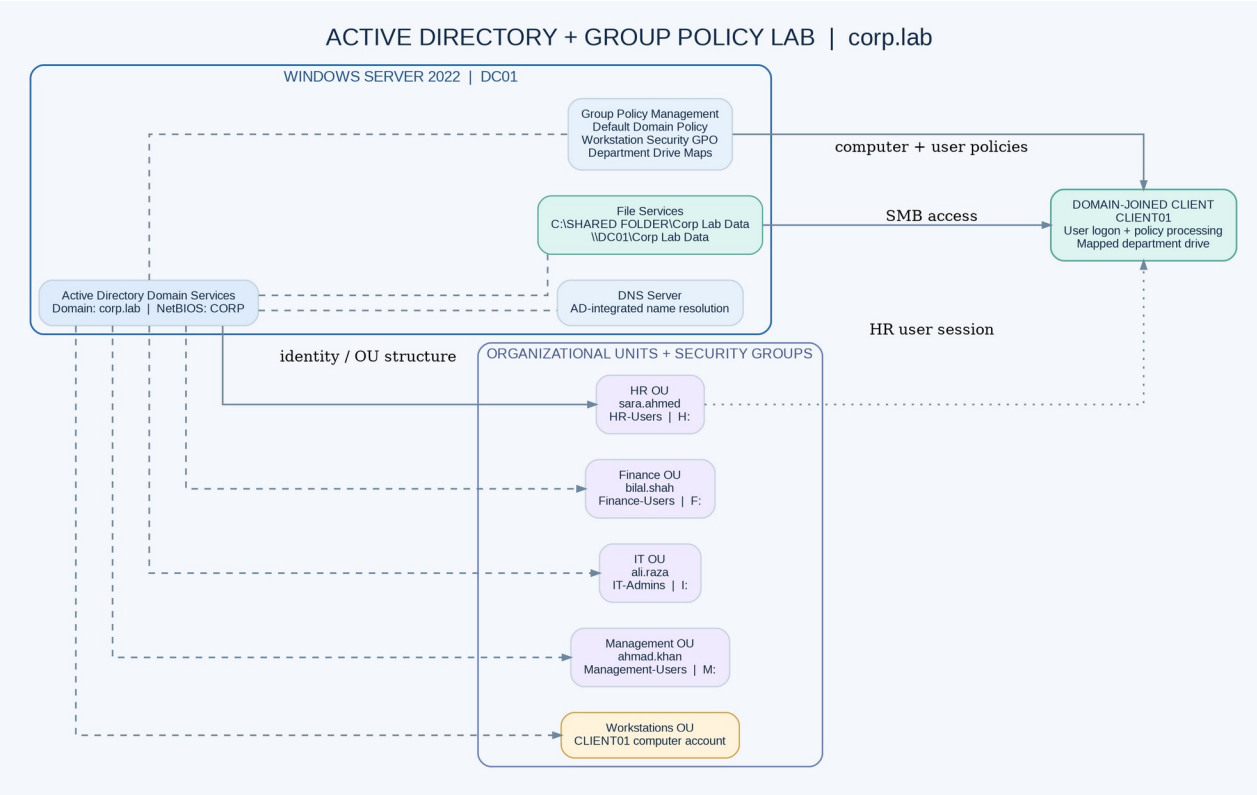


Figure 1. Logical domain, identity, file-service, and client-policy relationships.

3. Active Directory Design

The domain is organized by department. Departmental user accounts are placed in their respective OUs so user-based GPO links can be processed through the OU hierarchy.

Organizational Unit	Representative account / object	Security group
Management	ahmad.khan	Management-Users
IT	ali.raza	IT-Admins
HR	sara.ahmed	HR-Users
Finance	bilal.shah	Finance-Users
Workstations	CLIENT01 computer account	N/A

Design notes

- Use security groups to assign resource permissions rather than assigning permissions individually to each user.
- Keep user objects in the correct department OU. A GPO linked to an OU applies based on user/computer object location and security filtering, not merely because a user belongs to a matching group.
- Keep workstation computer accounts in the Workstations OU so computer-based policy scope can be controlled separately from user-based restrictions.

4. File Services and Department Drive Maps

The share root is hosted at C:\SHARED FOLDER\Corp Lab Data and published as Corp Lab Data. The department folders are Finance, HR DATA, IT, and Management.

Department	Folder / UNC target	Drive	Label
HR	\\DC01\Corp Lab Data\HR DATA	H:	HR Shared Drive
Finance	\\DC01\Corp Lab Data\Finance	F:	Finance Shared Drive
IT	\\DC01\Corp Lab Data\IT	I:	IT Shared Drive
Management	\\DC01\Corp Lab Data\Management	M:	Management Shared Drive

Drive-map design

The maps use User Configuration > Preferences > Windows Settings > Drive Maps. The configured preference uses the Update action, the department UNC path, a fixed drive letter, a descriptive label, and Reconnect. The HR H: drive is visible in the supplied client screenshot under sara.ahmed.

Permissions review The supplied share-permission screenshot shows Everyone with Full Control. The NTFS folder ACLs should remain the authoritative department boundary, but broad share permissions should be narrowed and the combined effective access tested.

5. Domain Account Policies

The Default Domain Policy screenshot records the following settings. They are reproduced as observed, not presented as a recommended universal baseline.

Password policy setting	Captured value
Enforce password history	24 passwords remembered
Maximum password age	50 days
Minimum password age	1 day
Minimum password length	5 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

Account lockout policy

Account lockout setting	Captured value
Account lockout threshold	10 invalid logon attempts
Account lockout duration	5 minutes
Reset account lockout counter after	5 minutes
Allow Administrator account lockout	Enabled

Baseline improvement The captured minimum password length is 5 characters. Raise it to at least 12 characters (14 is a stronger lab target), then update the evidence and confirm the resulting password policy.

6. Group Policy Inventory

GPO	Link / target	Main purpose
Default Domain Policy	corp.lab domain	Password and account-lockout policy
GPO-HR-Drive-Mapping	HR OU	H: mapped to HR DATA
GPO-Finance-Drive-Mapping	Finance OU	F: mapped to Finance
GPO-IT-Drive-Mapping	IT OU	I: mapped to IT
GPO-Management-Drive-Mapping	Management OU	M: mapped to Management
GPO-HR-Desktop-Restrictions	HR OU	Prohibit access to Control Panel and PC settings
GPO-Domain-Workstation-Security	Linked at domain root in lab	Central removable-media and user/workstation restrictions

Central workstation restrictions

Policy / control	Recorded state	Evidence status
All Removable Storage classes: Deny all access	Enabled	Policy screen captured; USB client test not shown
CD and DVD: Deny read / write access	Configured	Client D: access denied captured
Turn off AutoPlay	Enabled; All drives	Policy screen captured
Prevent access to Command Prompt	Enabled	Policy screen captured; client test not shown
Prevent access to Registry Editor	Configured in working session	Separate client test not shown
Prohibit access to Control Panel and PC settings	Enabled	Policy screen and restriction message captured
Settings Page Visibility	Selected Settings pages hidden	Configured in working session; separate Settings test not shown
Prevent changing proxy settings	Enabled (legacy IE template)	Policy screen captured
PowerShell executable restriction	Configured in working session	Client test not shown
Remove Run command from Start	Configured in working session	Client test not shown

7. Validation Summary

The evidence package includes screenshots of the Active Directory structure, share creation and permissions, account policies, drive-map configuration, security restrictions, and the HR client session.

Validation item	Evidence-based result
Active Directory structure	corp.lab with departmental OUs and groups is visible.
SMB share	Corp Lab Data share is visible; broad Everyone share permission needs remediation.
Password / lockout settings	Configuration screens captured; values listed in this report.
HR domain logon	sara.ahmed session visible on CLIENT01.
HR H: drive	Mapped HR DATA network location visible in File Explorer.
Control Panel restriction	Windows Restrictions message captured.
CD/DVD access	D: access denied message captured.
USB / CMD / Registry / PowerShell / Run	Configuration evidence exists for some controls, but client-side pass evidence is not included.
Finance / IT / Management drive maps	GPO configuration screens are included; individual client tests are not shown.

Reporting rule Use “configured” for a setting visible in Group Policy Management Editor and “passed” only for an observed client-side behavior. This distinction makes the evidence accurate and defensible.

8. Security Review and Required Remediation

Critical: Always install with elevated privileges

One supplied Windows Installer policy screenshot shows Always install with elevated privileges set to Enabled. This setting can allow Windows Installer to run packages with elevated system permissions; it is not the same as blocking MSI installations.

- Set Always install with elevated privileges to Disabled in Computer Configuration > Policies > Administrative Templates > Windows Components > Windows Installer.
- Set the same policy to Disabled in User Configuration > Policies > Administrative Templates > Windows Components > Windows Installer.
- If the intended restriction is to prevent users from installing MSI packages, configure the distinct Disable Windows Installer policy with the intended option and verify the behavior.

Other improvements

- Increase the minimum password length from 5 characters to at least 12 (14 is a stronger lab target).
- Replace Everyone: Full Control at the SMB share level with narrower authenticated access; retain Administrators Full Control and enforce department-specific NTFS ACLs.
- Review the domain-root link of GPO-Domain-Workstation-Security. Consider separating computer controls for the Workstations OU from user restrictions for standard department users. Avoid unintended restrictions on domain controllers, administrators, and service accounts.
- Review the 10-attempt / 5-minute account lockout values and document the chosen threshold, lockout duration, and recovery approach.

Microsoft documents the risks of AlwaysInstallElevated and recommends avoiding it. References: Microsoft Learn, Windows Installer Best Practices; Microsoft Learn, Machine Policies.

9. Operational Checklist and Next Steps

1. Correct AlwaysInstallElevated in both User and Computer Configuration.
2. Increase the domain minimum password length and re-capture the policy screenshot.
3. Tighten share-level permissions and verify HR, Finance, IT, and Management NTFS folder access.
4. Review GPO link scope and ensure standard users are targeted without unnecessarily restricting administrators or domain controllers.
5. Test USB, Command Prompt, Registry Editor, PowerShell, Run, and Settings separately on CLIENT01.
6. Sign in as bilal.shah, ali.raza, and ahmad.khan to verify F:, I:, and M: respectively.
7. Capture a final screenshot for every passed test and update Validation-Results.md.

10. Evidence Index

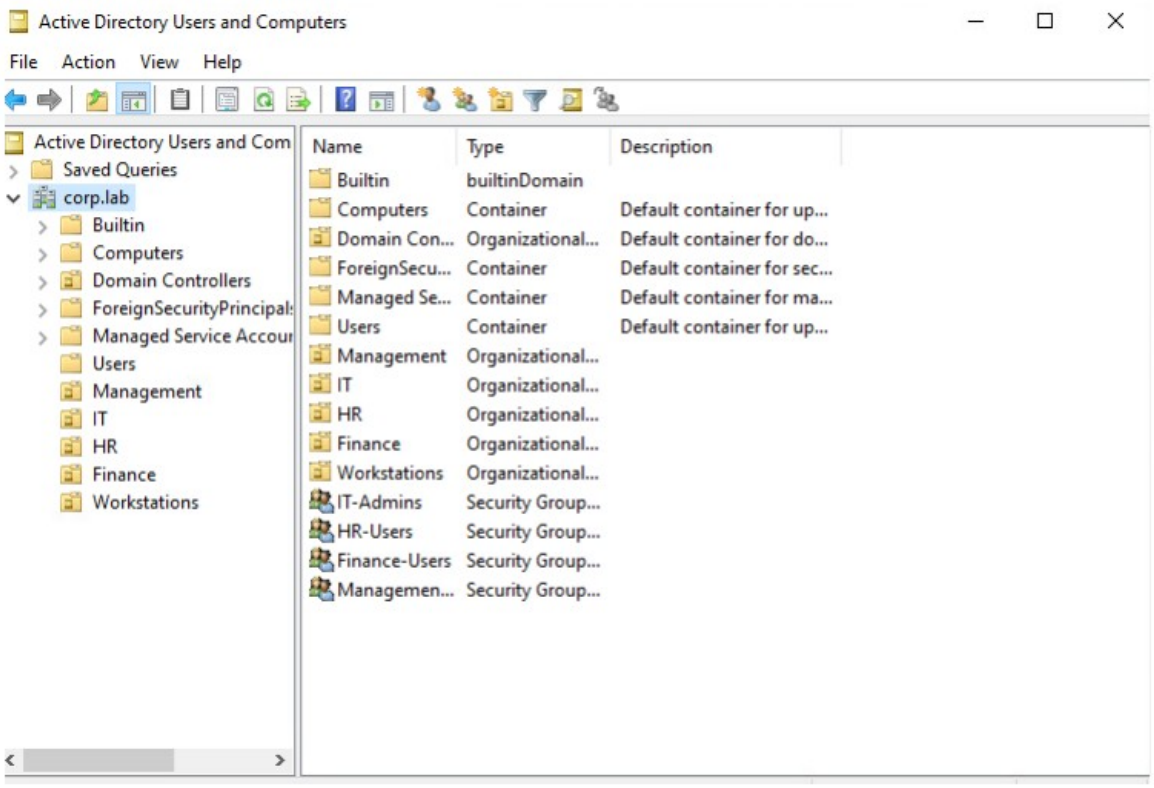
Screenshot	Content
01	Server Manager and network details
02	New forest / corp.lab deployment configuration
03	Active Directory OUs and security groups
04	Corp Lab Data share creation
05	Share permissions and folder access settings
06	Default Domain Policy password settings
07	Account lockout policy and drive-map configuration
08	Department drive-map GPOs
09	Removable-storage and AutoPlay settings
10	Command Prompt, Control Panel, and Windows Installer policies
11	sara.ahmed logon, H: drive, and proxy policy
12	Control Panel restriction and CD/DVD access denied

Appendix A - Configuration and Test Evidence

The following pages reproduce the supplied reference captures. Captions identify what each page supports; they do not imply that every policy shown was independently tested.

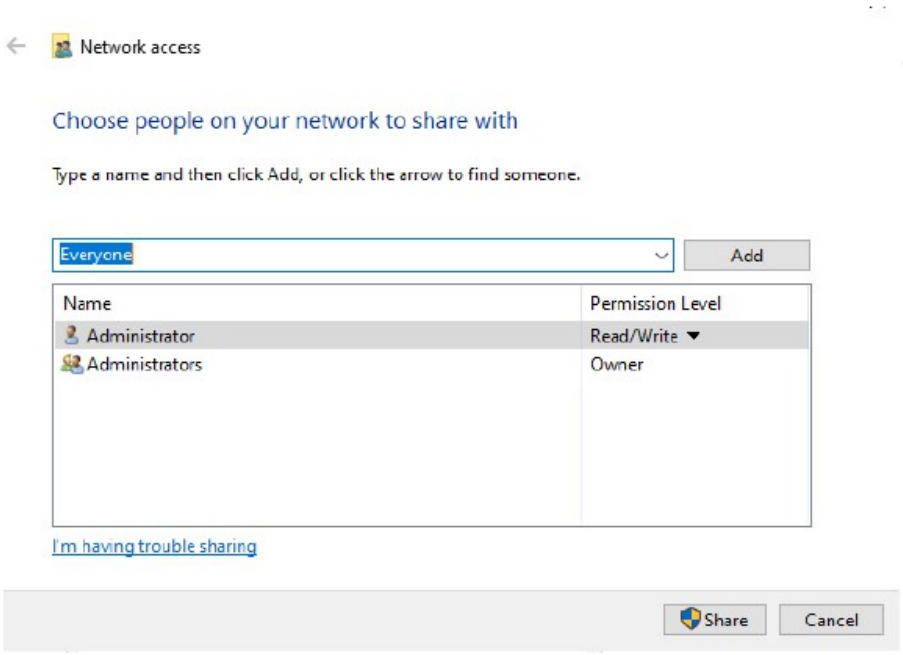
Appendix A.1 - Active Directory structure

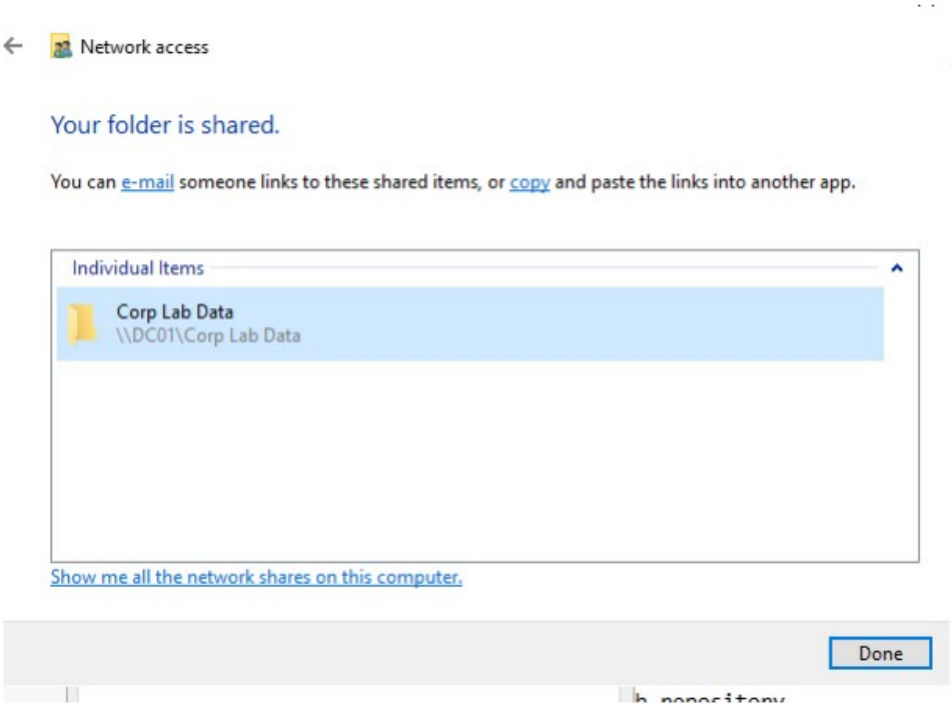
Departmental OUs and the HR, IT, Finance, and Management security groups are visible.



Appendix A.2 - SMB share creation

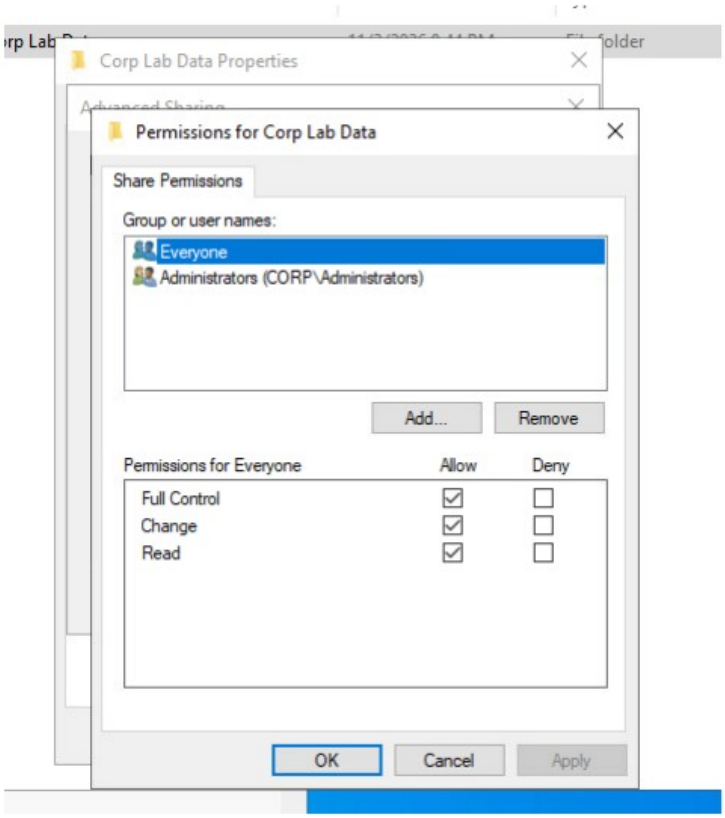
The Corp Lab Data share is created and the network UNC is shown.

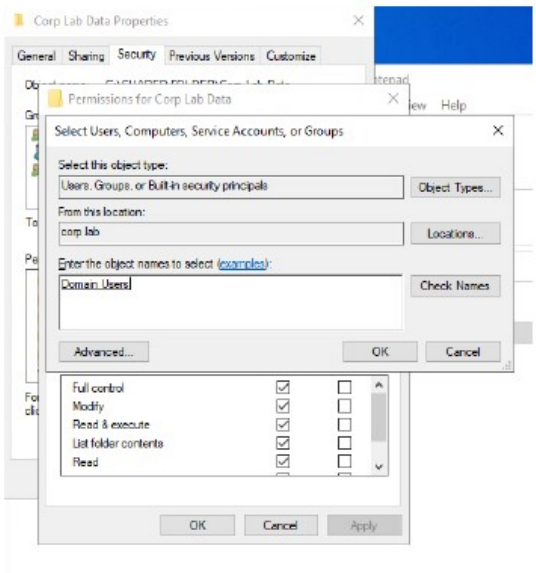




Appendix A.3 - Share permissions

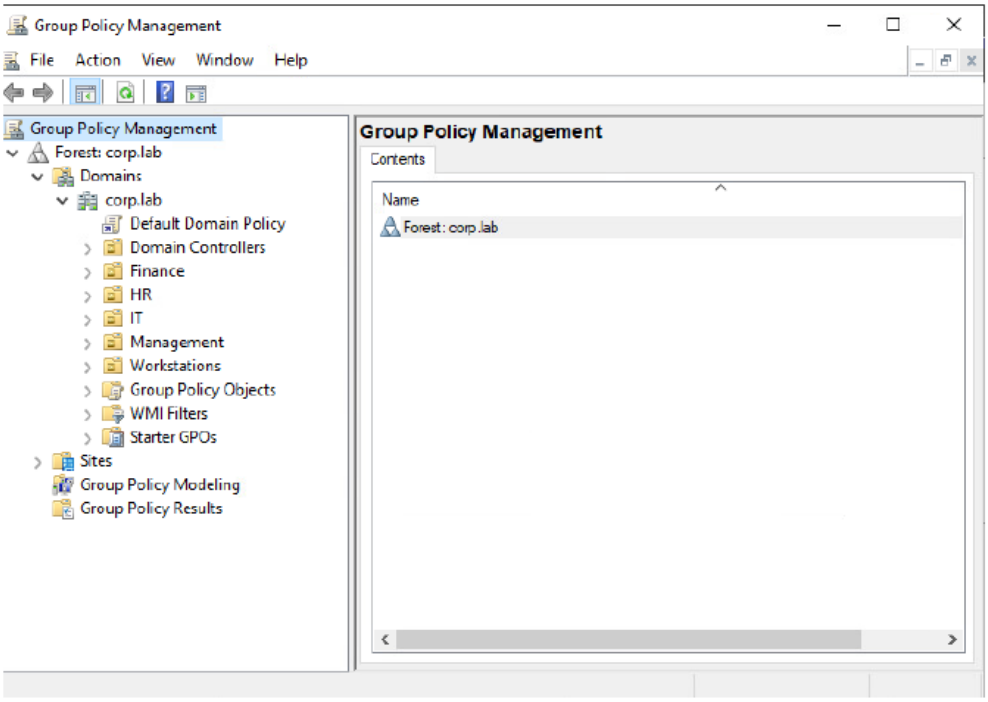
The screenshot records Everyone and Administrators share permissions. Everyone Full Control is flagged for remediation.

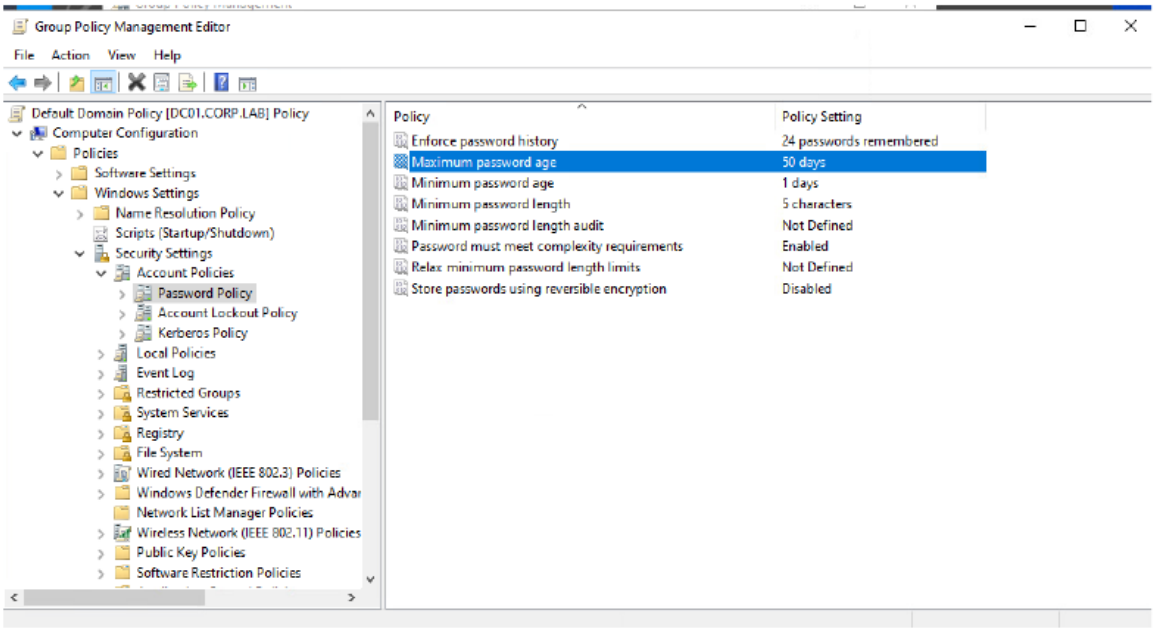




Appendix A.4 - Domain password policy

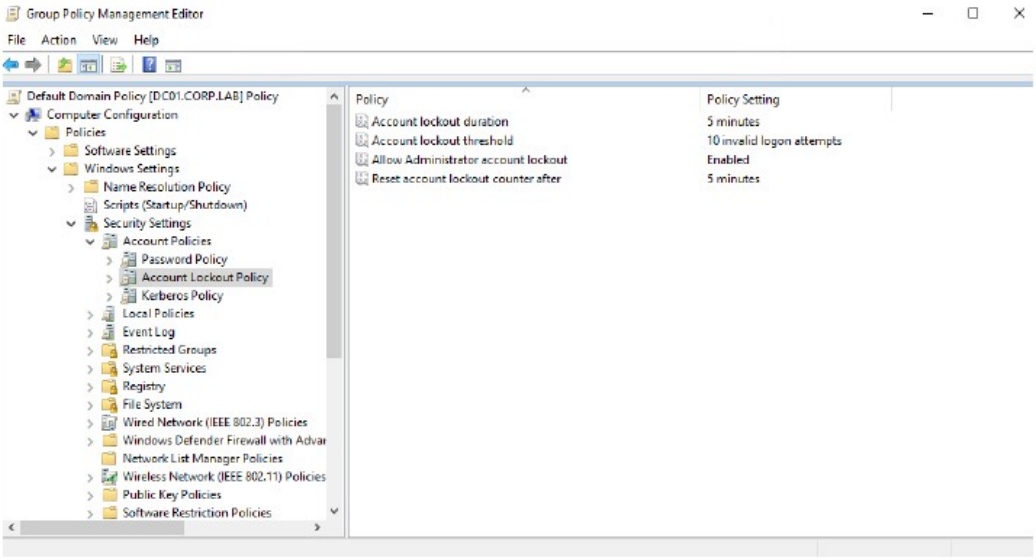
Captured password history, age, minimum length, complexity, and reversible-encryption settings.

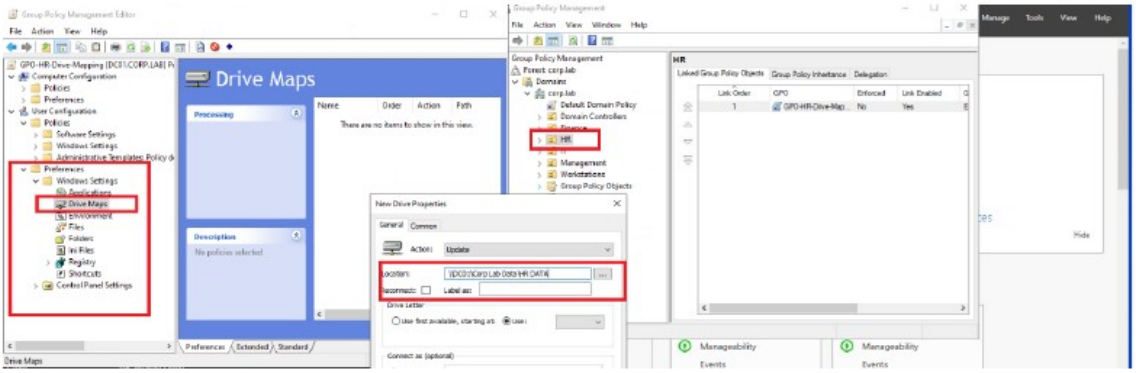


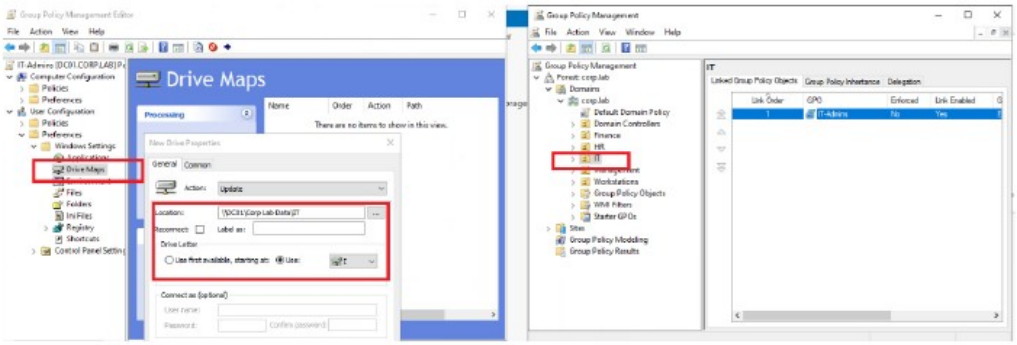


Appendix A.5 - Account lockout and HR drive-map setup

Captured lockout values and HR drive-map preference configuration.

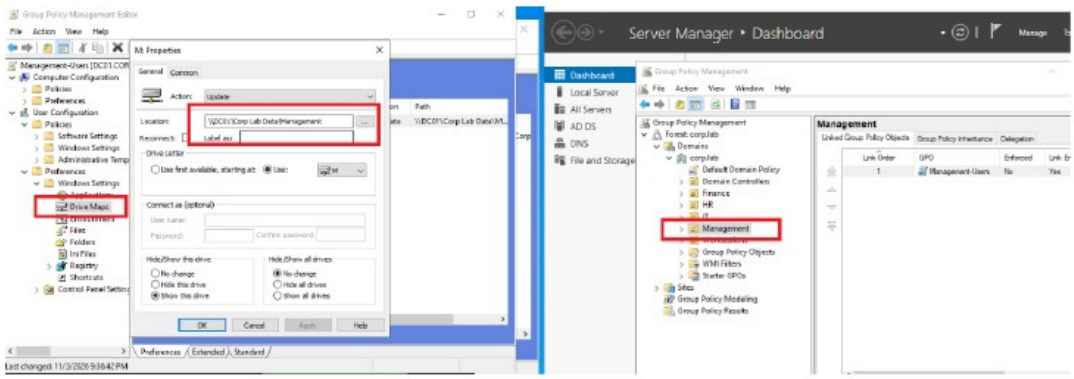


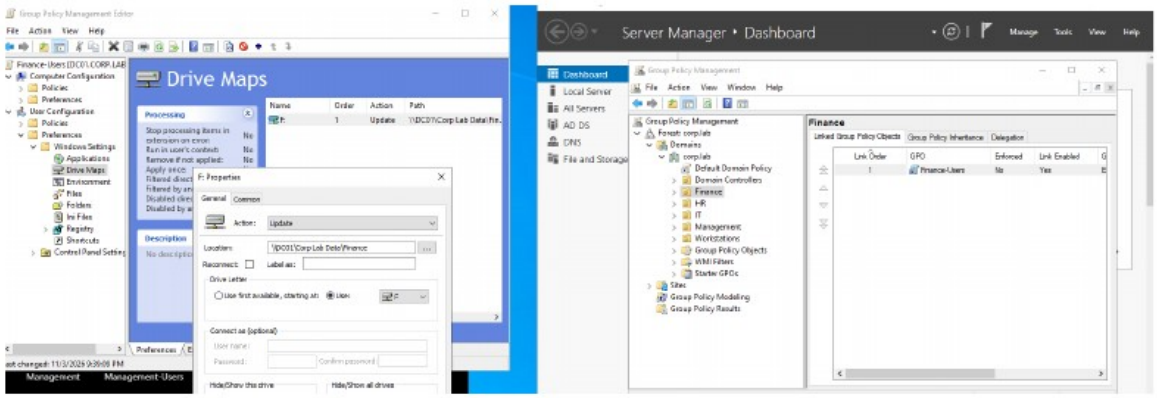


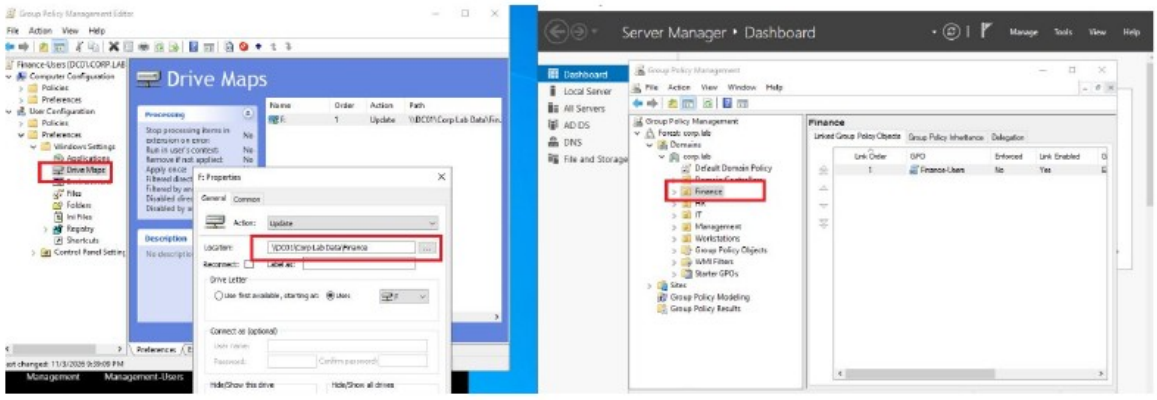


Appendix A.6 - Department drive-map GPOs

Management and Finance drive-map configuration and OU links.

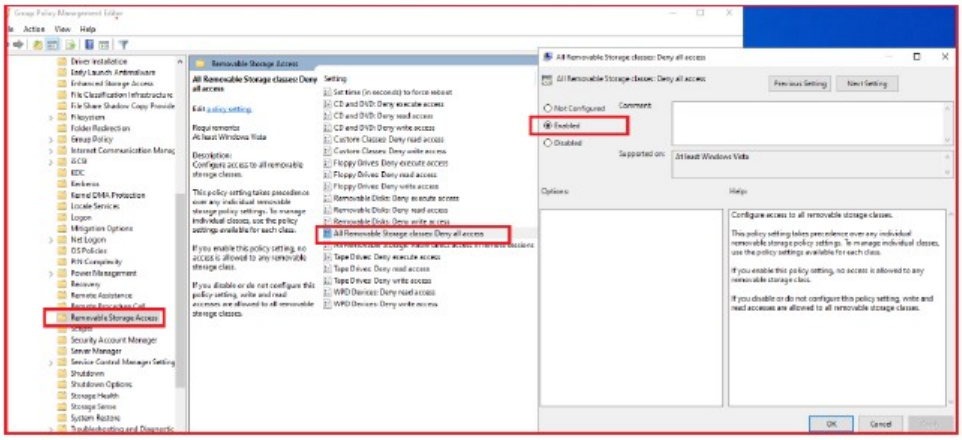


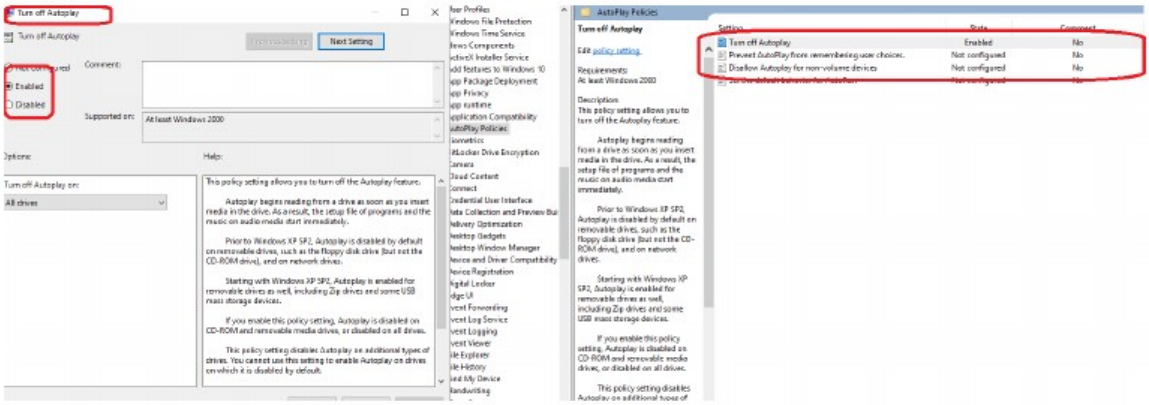


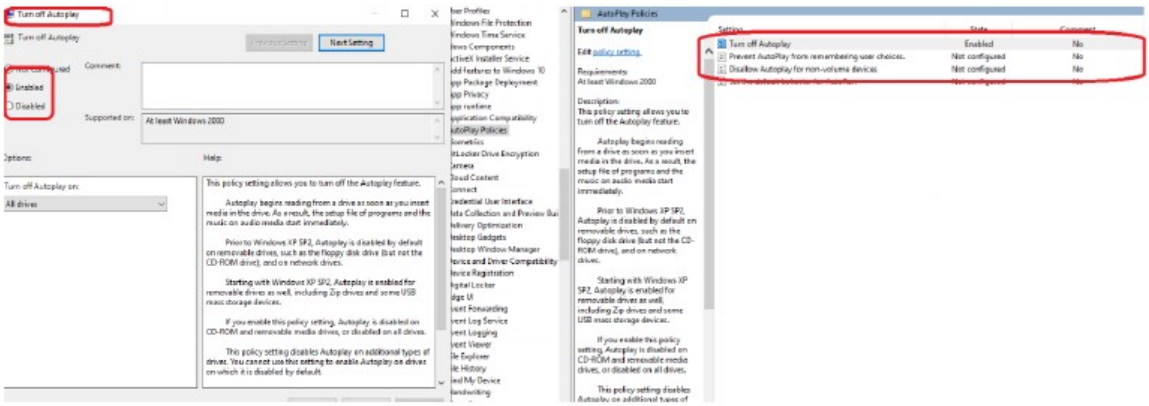


Appendix A.7 - Removable storage and AutoPlay

All removable storage deny access and AutoPlay disabled for all drives.

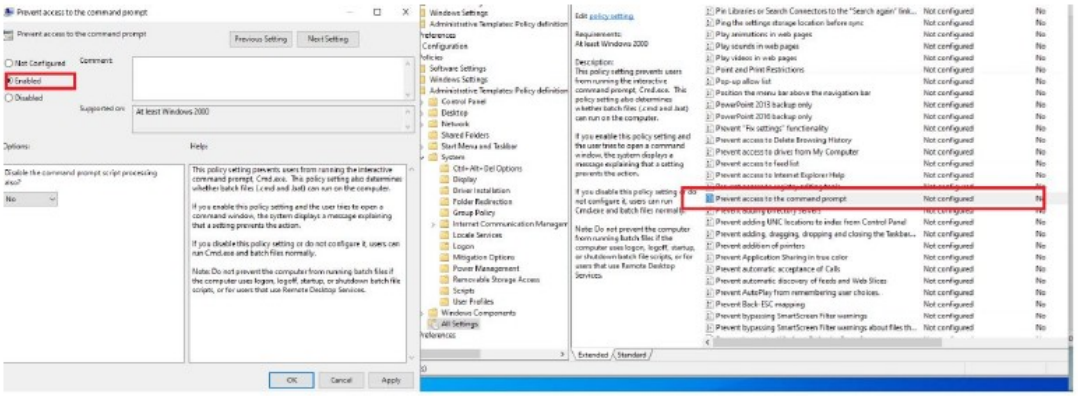


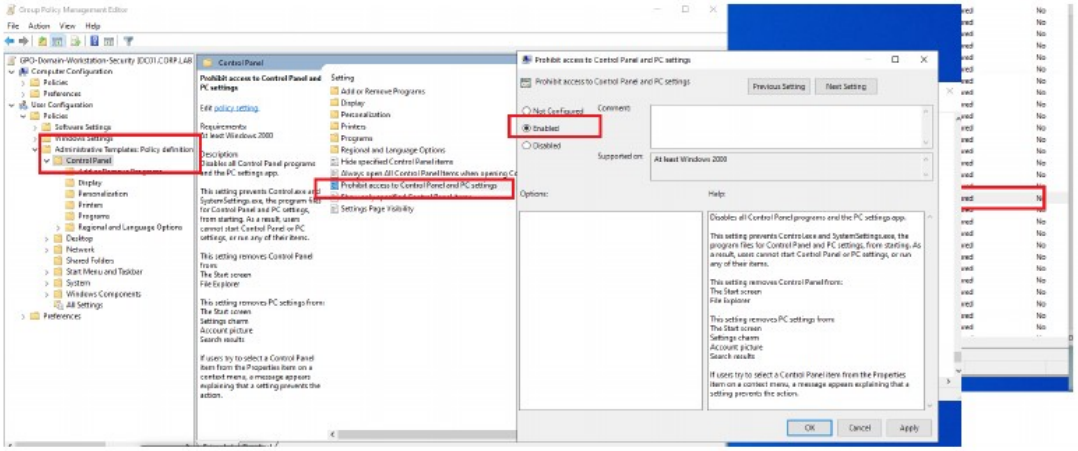


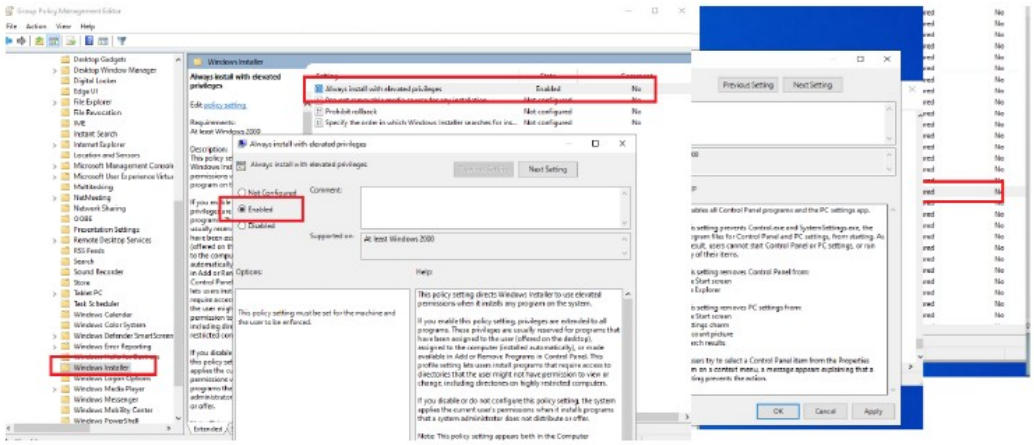


Appendix A.8 - Command Prompt, Control Panel, and Windows Installer

Shows restriction settings and the enabled Always install with elevated privileges setting that requires correction.

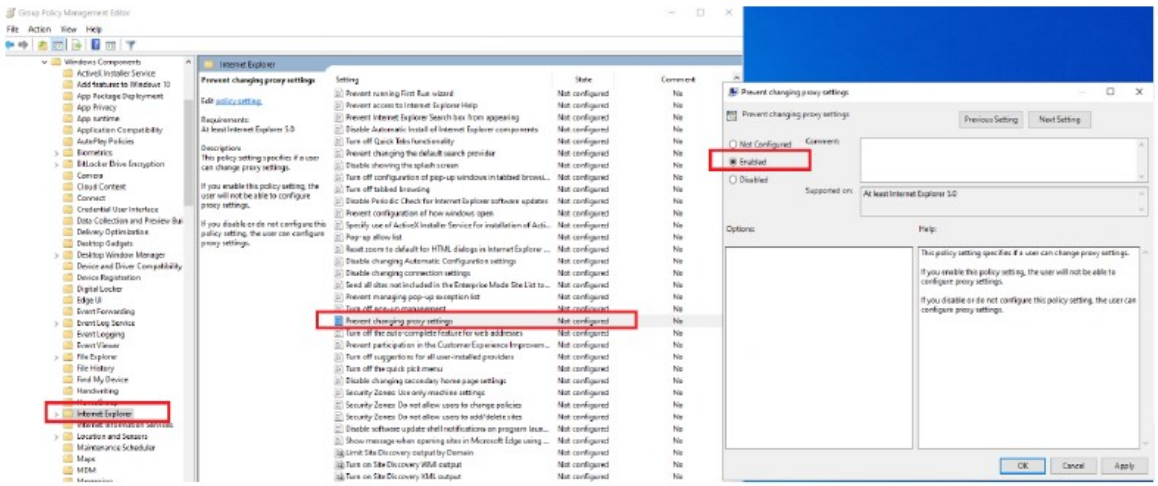


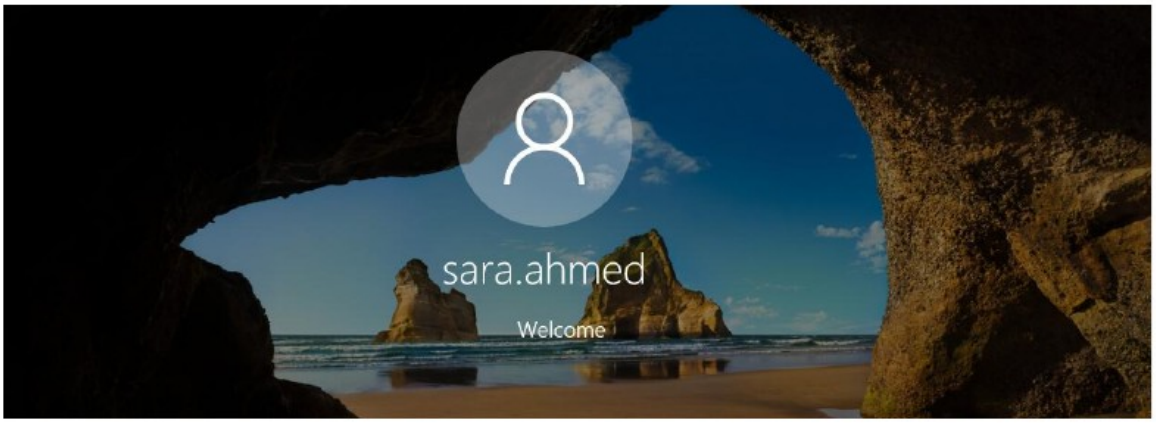


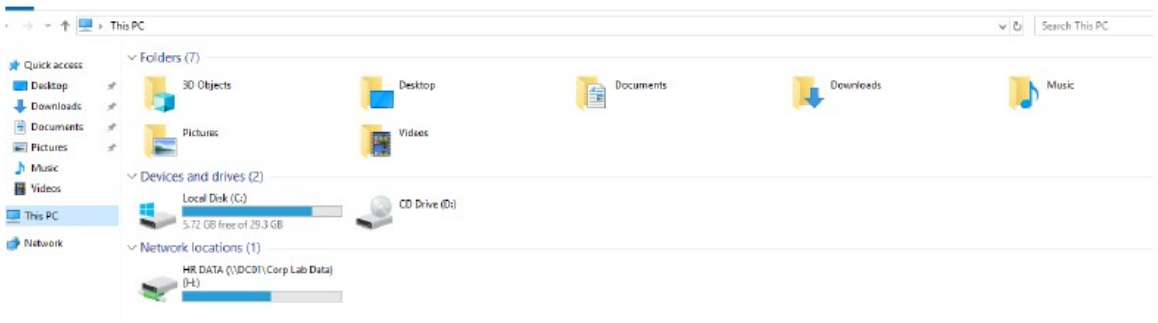


Appendix A.9 - HR session and H: drive

sara.ahmed session and the mapped HR DATA H: network location are visible.







Appendix A.10 - Client restriction outcomes

Windows Restrictions message and CD/DVD D: access denied are captured.

